Ajinkya Awari

London, UK · ajinkya.awari.26@ucl.ac.uk · linkedin.com/in/ajinkya-awari-641114249 · github.com/ajinkya-awari

EDUCATION

University College London (UCL), London, UK 2026 – 2027
MSc Artificial Intelligence for Biomedicine and Healthcare

Savitribai Phule Pune University, Pune, India 2019 – 2023
BE Computer Science · CGPA 8.25 · Vice President, Technology Club

SELECTED RESEARCH PROJECTS (full portfolio of 21 projects: github.com/ajinkya-awari)

- **AIXI-Complexity Benchmark** – compared 5 agent types on 20 frozen finite-state tasks; 81 tests, CI passing; no difficulty proxy remained significant after Holm correction.
- **FedScope** – federated learning (FedAvg, FedProx, FedNova) against a centralized baseline; F1 0.78–0.80 over 3 seeds.
- **GraphGPS-Upgrade** – GraphGPS vs GCN and GIN under one frozen 30-epoch budget; test ROC-AUC 0.748 vs 0.698 vs 0.693 (3 seeds).
- **AFBind** – AlphaFold2-guided protein–ligand affinity benchmark on BindingDB Ki; leakage-safe cross-target split; ECFP6 baseline Pearson r = 0.4765; 37 tests.
- **ESM2-Protein-Fitness** – protein mutation fitness prediction from ESM2 embeddings; 112 tests; Kaggle run complete.
- **PrivacyFL-DP** (in progress) – differentially private federated learning on NIH ChestX-ray14; 1 of 6 operating points measured (ε = 0.096, macro-AUC 0.497, chance level), five pending.

PUBLICATIONS

- V. Paithankar, **A. Awari**, A. Raskar, S. Patil, N. Jamdar. "Plant Disease Detection using Machine Learning." *IJARSCT*, Vol. 3, Issue 2, April 2023. DOI 10.48175/IJARSCT-9156.
- V. Paithankar, **A. Awari**, A. Raskar, S. Patil, N. Jamdar. "Plant Disease Detection using Machine Learning." *IJARSCT*, Vol. 3, Issue 4, April 2023. DOI 10.48175/IJARSCT-9297.

EXPERIENCE

The Leadership 30 – Senior Data Analyst July 2024 – Present

- Transforming Education in Maharashtra (SDG 4): led a data-driven initiative with the Maharashtra government on a three-month project to assess learning gaps in public schools; developed intervention strategies that boosted student engagement by 15% and improved literacy outcomes for 2,000+ students in rural districts.
- Advancing Clean Energy Access (SDG 7): designed a cost-benefit analysis model for renewable energy projects in rural Maharashtra, supporting policy recommendations for solar micro grids; the initiative led to a 10% increase in clean energy adoption, benefiting 800+ households and reducing annual carbon emissions by 1,500 tons.
- Disaster Resilience & AI-Driven Risk Mitigation: built an AI-powered flood forecasting system with local agencies, improving emergency response times by 12% in flood-prone districts of Maharashtra; assessed hardware and software requirements for such systems.

Byju's Think & Learn Pvt. Ltd, India – Software Developer July 2023 – July 2024

- Optimized lead management automation by integrating LeadSquared, reducing manual effort by 4+ hours per week and boosting lead conversion rates by 10%, contributing to $300K+ in additional revenue.
- Engineered a scalable platform capable of handling 300,000+ concurrent users by optimizing backend architecture and database queries, reducing latency by 15% and ensuring 99.7% uptime during peak enrolments.
- Developed a predictive lead scoring system in Python, improving lead-to-customer conversion by 12% and driving an 8% increase in quarterly revenue.
- Automated user onboarding workflows through Python-powered CRM integration, reducing onboarding time by 25% and errors by 30%.

INTERNSHIPS

Humanity First Foundation – AI & Data Science Intern Jan – Mar 2023

- Helped develop a heatwave prediction model for Vidarbha, supporting early health advisories to reduce heatstroke incidents.

Adani Group – Cyber Security & Data Analytics Intern Jun – Aug 2022

- Assisted in analysing cyber security risks for Maharashtra's power grid and supported an AI-based anomaly detection system for IoT security monitoring.

EARLIER PROJECTS

- **Customer Churn Prediction** (Jul – Oct 2022) – Random Forest churn forecasting to support retention strategies.
- **ETAMAX fest website** (Mar – Apr 2022) – design lead for a team of 7+; over three days the site drew 10,000+ users for registrations and exploration.

SKILLS

Machine learning: Python, PyTorch, PyTorch Geometric, Hugging Face, scikit-learn, Opacus, Kaggle, pytest.
Core: Python, C, C++, data structures and algorithms, DBMS, computer networks, Hadoop. **Web:** HTML, CSS, Bootstrap. **Other:** Agile methodology, operations management, content writing.